

WEEK – 4

Introduction to Classes, objects, constructors, methods, parameter passing, overloading constructors and methods.

Sample Program:

3. Classes and Objects CO1

Design a “farm animals” java application with the details of animals like cow, pig, horse.

Consider the following details like where they stay, what they eat, the sound they make by using classes and objects.

Algorithm

1. **Start** the program.
2. Define a class named Animals.
3. Declare four data members:
 - name
 - place
 - food
 - sound
4. Create a constructor Animals() that initializes these four variables.
5. Define the display() method to print the animal's details.
6. Define the FarmHouse class containing the main() method.
7. Create an Animals object named cow with:
 - Name = cow
 - Place = coewhed
 - Food = grass
 - Sound = aaaaaaa
8. Create an Animals object named pig with:
 - Name = pig
 - Place = pigshed
 - Food = grass
 - Sound = ooooo
9. Create an Animals object named horse with:
 - Name = horse

- Place = shed
- Food = grass
- Sound = hmmm

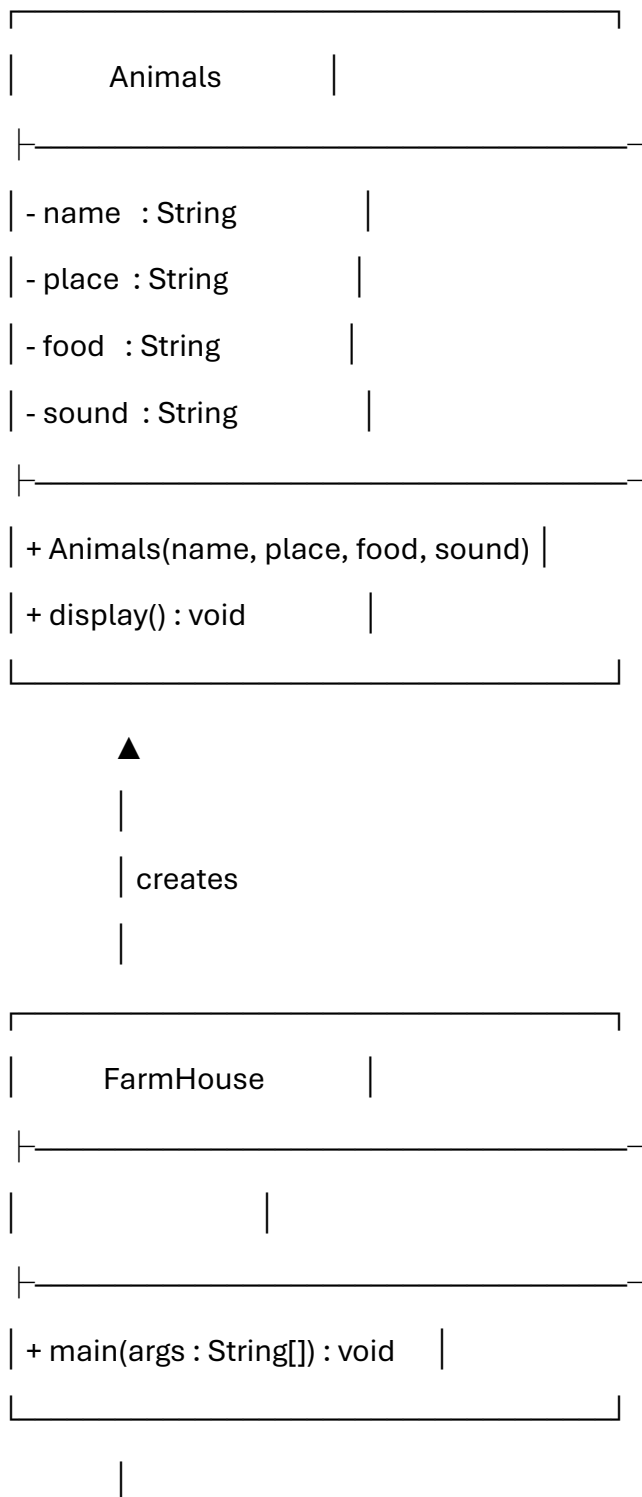
10. Call `display()` for the cow object.

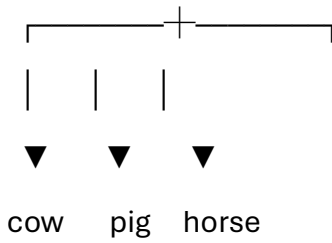
11. Call `display()` for the pig object.

12. Call `display()` for the horse object.

13. **Stop** the program.

UML Class Diagram





: Animals : Animals : Animals

4.CODE:

```
class Animals{
```

```
    String name;
```

```
    String place;
```

```
    String food;
```

```
    String sound;
```

```
    Animals(String name,String place,String food,String sound){
```

```
        this.name=name;
```

```
        this.place=place;
```

```
        this.food=food;
```

```
        this.sound=sound;
```

```
    }
```

```
    void display(){
```

```
        System.out.println("the animal "+name+" stays in "+place+" and eats "+food+" and make the sound "+sound);
```

```
    }
```

```
}
```

```
public class FarmHouse {
```

```
    public static void main(String[] args) {
```

```
        Animals cow=new Animals("cow","coewhed","grass","aaaaaaa");
```

```
        Animals pig=new Animals("pig","pigshed","grass","ooooo");
```

```
        Animals horse=new Animals("horse","shed", "grass", "hmmmm");
```

```

cow.display();

pig.display();

horse.display();

}

}

```

OUTPUT:

```

PS C:\Users\anits-csm\Desktop\String Operations> & 'C:\Program Files\Java\jdk-25.0.3\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\anits-csm\AppData\Roaming\Code
\User\workspaceStorage\24466a991fa60fc1415fd50ee149d70a\redhat.java\jdt_ws\String Operations_e709c925\bin' 'FarmHouse'
the animal cow stays in coewhed and eats grass and make the sound aaaaaaa
the animal pig stays in pigshed and eats grass and make the sound oooooo
the animal horse stays in shed and eats grass and make the sound hmmm
PS C:\Users\anits-csm\Desktop\String Operations> c:; cd 'c:\Users\anits-csm\Desktop\String Operations'; & 'C:\Program Files\Java\jdk-25.0.3\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionM
essages' '-cp' 'C:\Users\anits-csm\AppData\Roaming\Code\User\workspaceStorage\24466a991fa60fc1415fd50ee149d70a\redhat.java\jdt_ws\String Operations_e709c925\bin' 'FarmHouse'
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the animal pig stays in pigshed and eats grass and make the sound oooooo
the animal horse stays in shed and eats grass and make the sound hmmm
PS C:\Users\anits-csm\Desktop\String Operations>

```